Algorithm 1: Stage 1 Multi-Channel Candidate Retrieval & Controlled Negative Sampling

Input: Customer $u \in U$, Transactions D ($t < T_{cutoff}$), Catalog I, Cohorts A, Ratio $\lambda_{neg} = 5$

Output: Balanced training dataset $T_u = \{(u, i, y)\}$ with label $y \in \{0, 1\}$

$C_u \leftarrow \emptyset$
$C_{repurchase} \leftarrow \{i \in I \mid \exists\ (u, i, t) \in D \wedge t \geq T_{cutoff} - 28d\}$
$C_{bestseller} \leftarrow \text{TopK}(\{i \in I \mid \text{sales_count}(i, [T_{cutoff} - 7d, T_{cutoff}])\}, K = 12)$
$C_{age} \leftarrow \text{TopK}(\{i \in I \mid \text{sales_by_cohort}(i, \text{age_group}(u))\}, K = 8)$
$C_{copurchase} \leftarrow \{j \in I \mid \exists\ i \in C_{repurchase} \wedge \text{co_purchased}(i, j) \geq 5\}$
$C_u \leftarrow \text{Truncate}(C_{repurchase} \cup C_{bestseller} \cup C_{age} \cup C_{copurchase}, C_{max} = 100)$
$Y_{true} \leftarrow \{i \in I \mid \exists\ (u, i, t) \in D \wedge t \in [T_{cutoff}, T_{cutoff} + 7d]\}$
$\text{Positives} \leftarrow \{(u, i, 1) \mid i \in (C_u \cap Y_{true})\}$
$N_{cand} \leftarrow C_u \setminus Y_{true}$
$k_{neg} \leftarrow \min(|N_{cand}|, \lambda_{neg} \times |\text{Positives}|)$
$\text{Negatives} \leftarrow \text{SampleWithoutReplacement}(N_{cand}, k_{neg})$
return $T_u \leftarrow \text{Positives} \cup \{(u, j, 0) \mid j \in \text{Negatives}\}$

Algorithm 2: Real-Time In-Session Dynamic Re-Ranking & Diversity Enforcement

Input: Target user u, Model candidates $\{(i, \hat{p}(u, i))\}$, Active session stream E_u, Length $K = 12$

Output: Final context-adapted recommendation list $R_u = [i_1, ..., i_K]$

$a \leftarrow 0_{|V|}$ // Initialize session anchor vector across category vocabulary V
for each interaction event $e = (i_e, \tau_e, t_e) \in E_u$ (ordered newest to oldest, max 50) do:
    $\Delta t \leftarrow (t_{now} - t_e)$ in days
    $w(e) \leftarrow \omega(\tau_e) \times 2^{-\Delta t / t_{1/2}}$ // Half-life exponential decay $t_{1/2} = 3.0$ days
    $a[\text{category}(i_e)] \leftarrow a[\text{category}(i_e)] + w(e)$
end for
$B \leftarrow \{i_e \mid \tau_e = \text{PURCHASED} \vee (\tau_e = \text{FEEDBACK} \wedge \text{rating}(e) \leq 2)\}$ // Hard suppression set
$C_{active} \leftarrow \{(i, \hat{p}(u, i)) \in C_u \mid i \notin B\}$
for each candidate item $(i, \hat{p}) \in C_{active}$ do:
    $\text{boost}(i) \leftarrow \sum_{c \in \text{categories}} a[c] \cdot \text{Similarity}(\text{category}(i), c)$
    $s_{final}(i) \leftarrow \hat{p} \times (1.0 + \max(0.0, \text{boost}(i)))$
end for
$S \leftarrow \text{SortDescending}(C_{active}, \text{key} = s_{final})$
$R_u \leftarrow [\,], \quad G_{seen} \leftarrow \emptyset$
for each $(i, s) \in S$ do:
    if $\text{product_code}(i) \notin G_{seen}$ then:
        $R_u \leftarrow R_u \oplus [i], \quad G_{seen} \leftarrow G_{seen} \cup \{\text{product_code}(i)\}$
    if $|R_u| = K$ then break
return R_u // Final re-ranked, diversified Top-12 recommendation sequence